

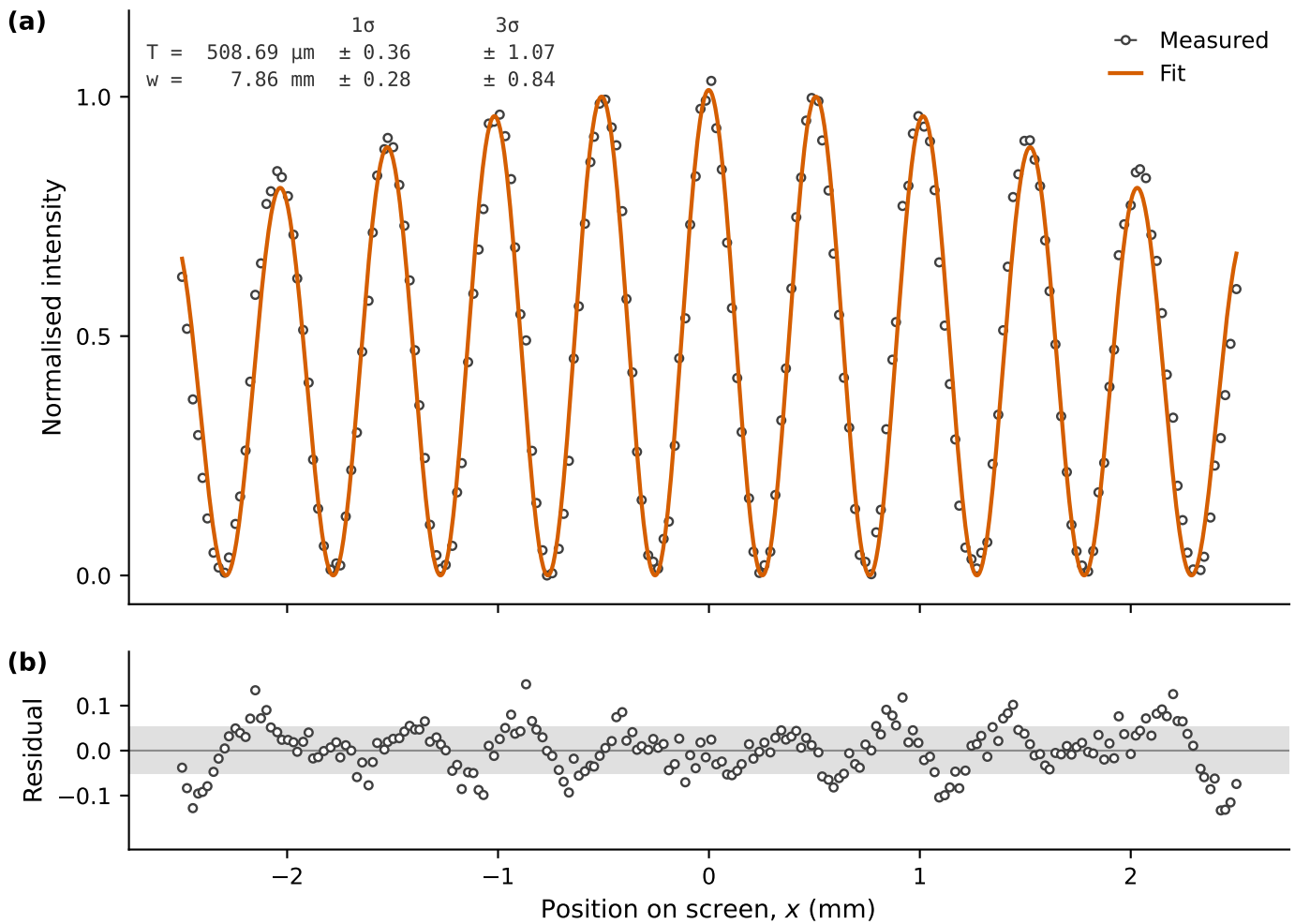


Figure 1: Two-slit interference profile and least-squares fit

(a) Normalised intensity measured as a function of position x on the observation screen (open circles; horizontal error bars, $\leq 8 \mu\text{m}$, are smaller than the markers), overlaid with a least-squares fit (orange line) to the two-slit model $I(x) = A \text{sinc}^2[(x - x_0)/w] \cos^2[\pi(x - x_0)/T]$, in which T is the fringe period, w the width of the diffraction envelope, x_0 the centre of the pattern and A the peak intensity. (b) Residuals of the same fit; the grey band spans ± 1 RMS residual (0.052). Nine maxima are fully resolved across the 5 mm field and are evenly spaced, giving $T = 508.69 \pm 0.36 \mu\text{m}$ (1σ ; 3σ interval 507.61–509.76 μm). The peak intensity falls from 1.03 at the centre to 0.85 at $x = \pm 2.0$ mm, described by a sinc^2 envelope of width $w = 7.86 \pm 0.28 \text{ mm}$ (1σ ; 3σ interval 7.02–8.70 mm). The residuals are unstructured within ± 0.05 over the central 4 mm but grow systematically beyond $|x| = 2.3$ mm. The constant fringe spacing confirms that the pattern is two-slit interference modulated by single-slit diffraction, and constrains the period to 0.07 %. The growth of the residuals at the edges of the field shows that the envelope is only weakly constrained over this range ($\cos^2$ and Gaussian envelopes fit equally well, RMS 0.052), so w should not be read as a precise slit width.